

Exalples

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Proposition 0.1. *Suppose that $\lambda \in X$, $\alpha_i \in \Phi_0^+$, $1 \leq i \leq k$, and $\beta, \gamma \in \Phi_1^+$. Let $w = s_{\alpha_k} s_{\alpha_{k-1}} \cdots s_{\alpha_1} \in \mathcal{W}$.*

- (1) *Suppose that $\langle \lambda, \alpha_1^\vee \rangle < 0$. Then $(P_\lambda : M_{s_{\alpha_1} \lambda}) > 0$.*
- (2) *Suppose that $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1} \lambda, \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w\lambda}) > 0$.*
- (3) *Suppose that $(\lambda, \beta) = 0$. Then $(P_\lambda : M_{\lambda+\beta}) > 0$.*
- (4) *Suppose that $(\lambda, \beta) = 0$ and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1} (\lambda + \beta), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta)}) > 0$.*
- (5) *Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$ and $\text{ht}(\beta) < \text{ht}(\gamma)$. Then $(P_\lambda : M_{\lambda+\beta+\gamma}) > 0$.*
- (6) *Suppose that $(\lambda, \beta) = (\lambda + \beta, \gamma) = 0$, $\text{ht}(\beta) < \text{ht}(\gamma)$, and $\langle s_{\alpha_{i-1}} \cdots s_{\alpha_1} (\lambda + \beta + \gamma), \alpha_i^\vee \rangle < 0$ for all $i \in \{1, 2, \dots, k\}$. Then $(P_\lambda : M_{w(\lambda+\beta+\gamma)}) > 0$.*

1 gl(2|2)

1.1 degree of atypicality = 1

Theorem 1.1. *Let $a, b, c \in \mathbb{Z}$ and $\lambda = (a, b|c, b)$. The projective objects P_λ in $\mathfrak{B}_{a|c}$ have the following Verma flag formulae.*

Proof. The Cases are as follows:

- (1) $a > b > c$

$$(1.1) \quad a > b + 1$$

Let $\mu := \lambda + (0, 1, 0, 0)$.

$$P_\mu = M_{a, b+1|c, b}$$

$$Pr_\lambda(P_\mu \otimes V_2) = M_{a, b|c, b} + M_{a, b+1|c, b+1}$$

All items belong to P_λ by [Theorem 0.1](#).

□